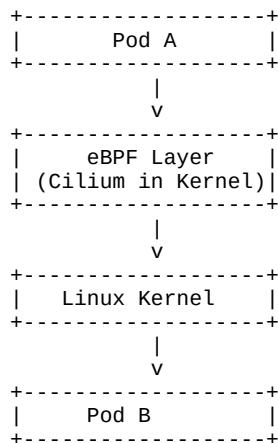


Cilium Kubernetes Networking Guide

1. What is Cilium?

Cilium is an advanced Kubernetes CNI (Container Network Interface) plugin based on eBPF (Extended Berkeley Packet Filter). It provides high-performance networking, security, and observability for container workloads.

2. Basic Architecture



3. Key Features

- L3/L4 Networking
- L7 (HTTP, gRPC, DNS) Policy Support
- Replaces kube-proxy (optional)
- High Performance using eBPF
- Built-in Observability with Hubble

4. Installation Steps (kubeadm Cluster)

Step 1: Remove existing CNI (if any)

```
kubectl delete -f flannel.yaml
```

Step 2: Install Cilium CLI

```
curl -LO https://github.com/cilium/cilium-cli/releases/latest/download/cilium-linux-amd64.tar.gz
tar xzvf cilium-linux-amd64.tar.gz
sudo mv cilium /usr/local/bin
```

Step 3: Install Cilium

```
cilium install
```

Step 4: Verify

```
cilium status  
kubectl get pods -n kube-system
```

5. Multi-AZ Cluster View

```
Region  
|  
+-- AZ-1a (Worker-1)  
|  
+-- AZ-1b (Worker-2)  
|  
+-- AZ-1c (Worker-3)
```

Cilium ensures secure and observable traffic between pods across nodes.